University of Applied Sciences

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The 12 Success Factors of Data Science

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Overview

The 12 Success Factors of Data Science

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2. The team - Interdisciplinary
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4. The development - from data hacker to software engineer
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6. The analysis methods - the big method case instead of "one-algo-fits-all"

The 12 Success Factors of Data Science

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12. Communication - Comprehensible messages and empathy instead of statistical gibberish

The 12 Success Factors of Data Science

1 - The Entry

Bottom-Up instead of Top-Down

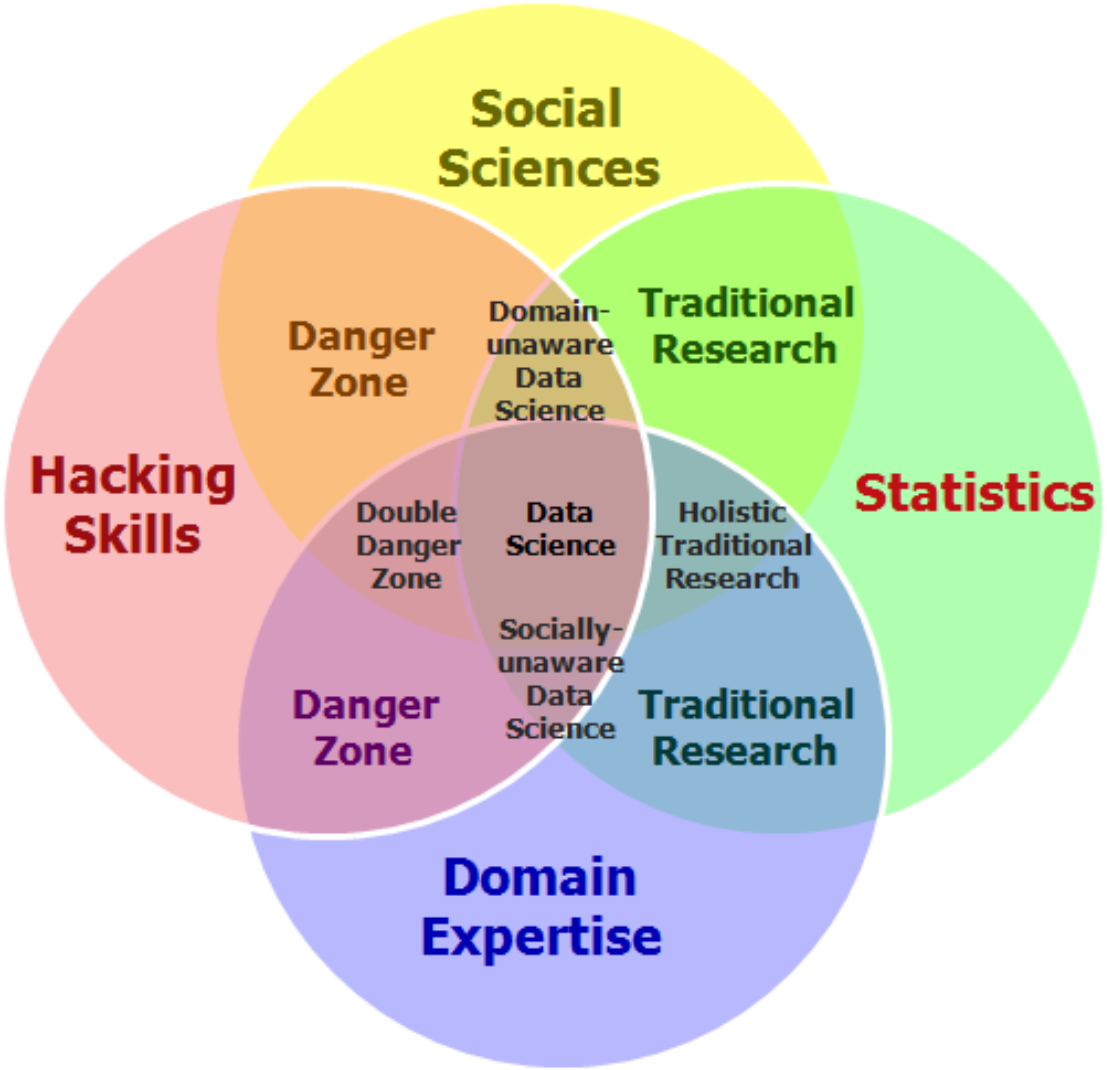
1 - The Entry

In many companies, the introduction of Data Science is a bottom-up process: Career starters and interns bring the skills they learned during their studies in the use of programming languages such as Python into the companies and achieve successes that surprise established employees.

2 - The Team

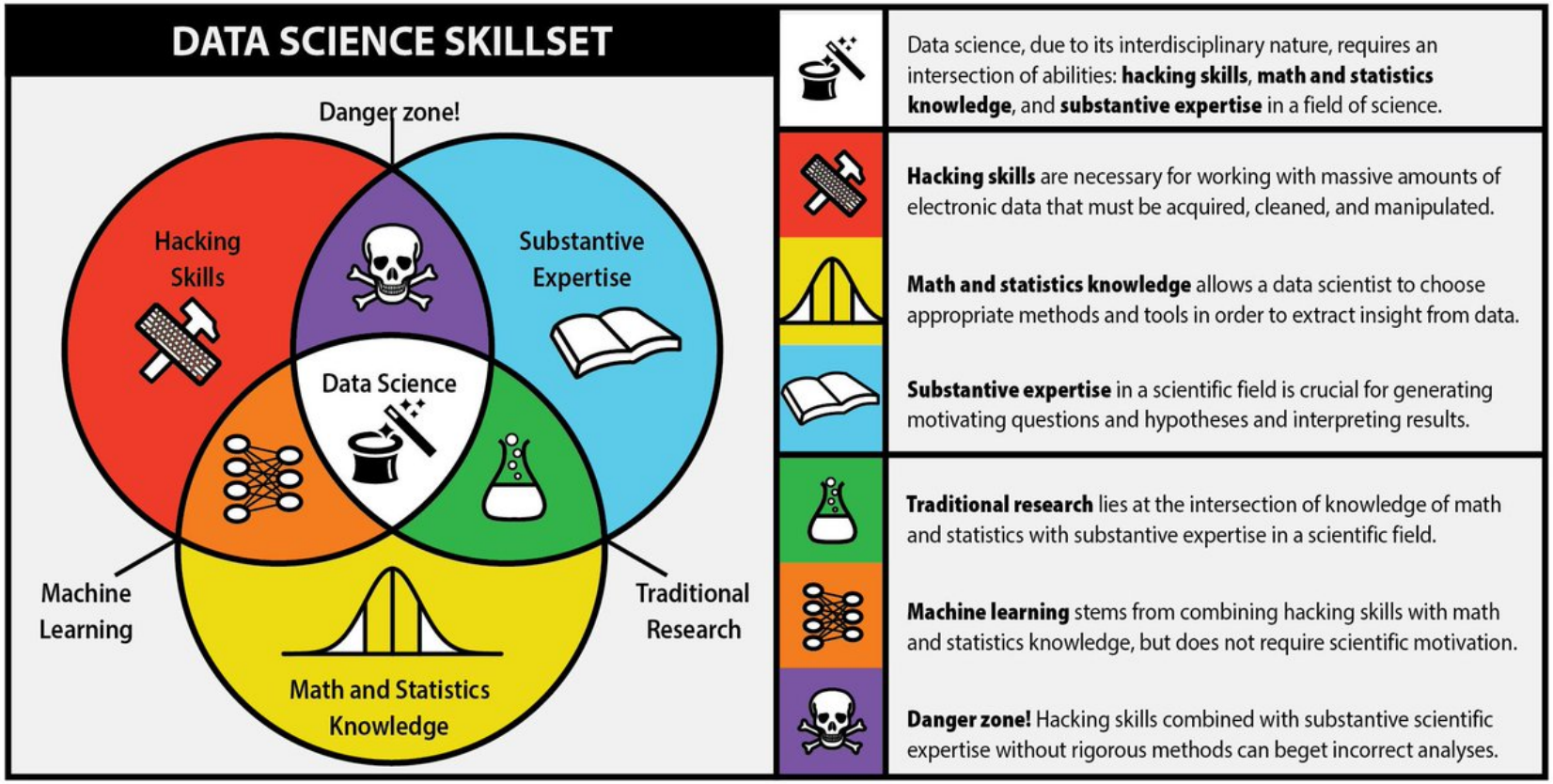
Interdisciplinary

2 - The Team



Source: KDnuggets.com

2 - The Team



Source: `twitter.com/mrdatascience/status/849234570163695616`

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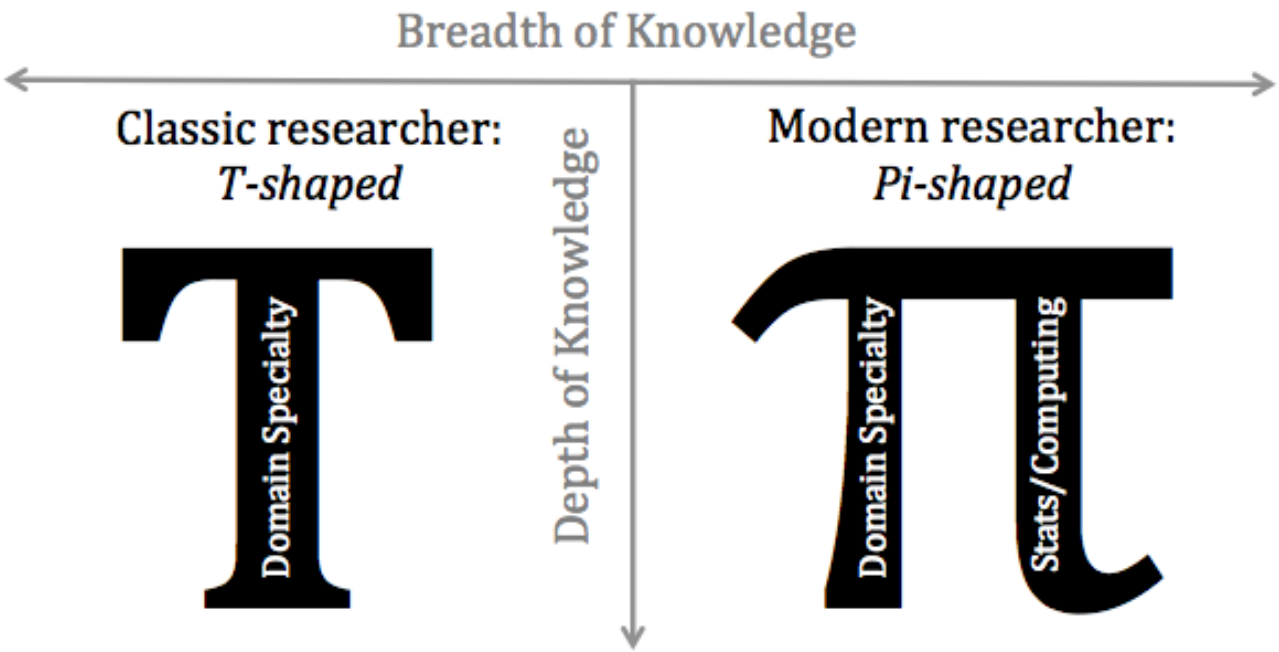
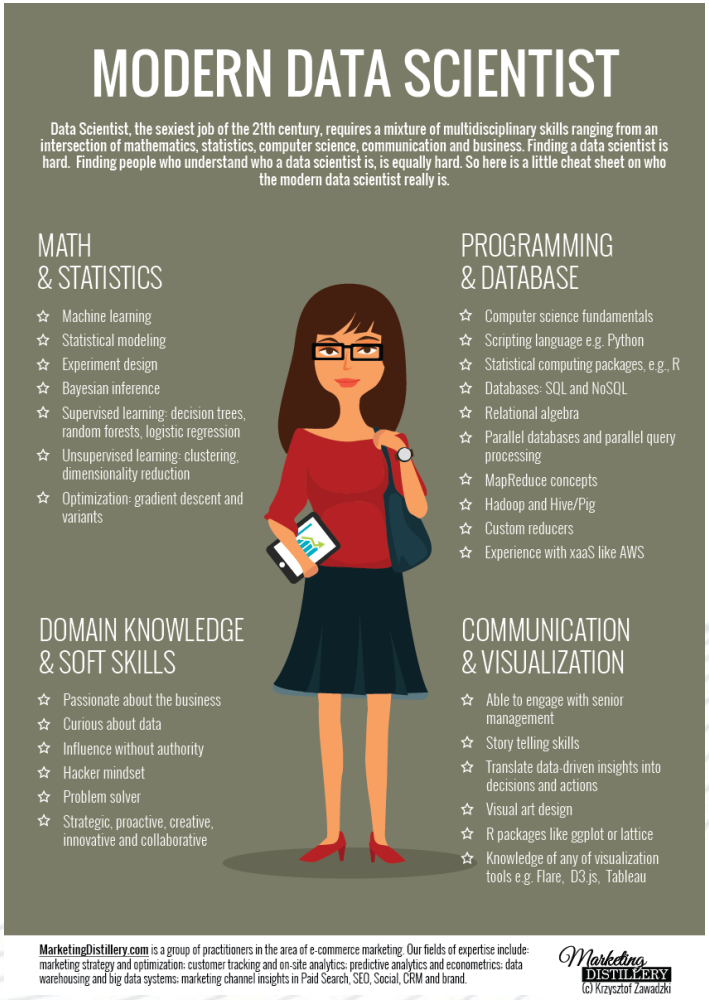


Source: Stefano Ceri: On the Big Impact of "Big Computer Science"/Drew Conway, CEO and founder of Alluvium

2 - The Team



Source: https://imgur.com/4ZBBvb0

2 - The Team

Statistics, computer science, mathematics, software development and specialist sciences with a statistical focus - Data Science is a broad field and stands for **interdisciplinary skills**.
These are combined with a detailed understanding of **company-specific business processes** and the distinct **competence to communicate complex issues** even to laymen.
Data Science is therefore one thing above all:

The performance of a **team** with different professional backgrounds and different skills.

3 - Further Education

Lifelong Learning at the Limit

3 - Further Education

The development of analysis methods, frameworks and technologies is rapid. In addition to the mathematical-statistical complexity that new approaches bring with them, there is the technical challenge. Data Science is often at the limit of what is technically feasible. Massively parallel systems or the shifting of computation from CPU to GPU requires in-depth architectural knowledge.

3 - Further Education

Data Science therefore stands in particular for lifelong learning. In order to be successful in the long term, data scientists **need the freedom** to familiarize themselves with new methods and procedures. This freedom must also be **reflected in project management**. If data scientists are deprived of this freedom through workloads in day-to-day business, they will gradually lose **touch with state-of-the-art technologies**.

4 - The Development

From Data Hacker to Software Engineer

4 - The Development

Data Scientists are rarely trained as software developers. Mostly they have developed their "hacking skills" on the side. The closer an analysis gets to the operational systems, the more important it is to apply the **structured approach of software development**.

4 - The Development

This includes

- code **versioning**,
- unit **testing**,
- performance **optimization**, and
- **standardized documentation**.

5 - The Toolset

Open Source at its Core

5 - The Toolset

The choice of the right tools depends both on the **specific application** and on the **preferences of the team**.

In general, however, it can be said that Data Science relies on **open source** at the core of the analysis toolset used.

5 - The Toolset

For example, the Data Science programming languages R and Python have more than 10,000 analysis packages. The growth in packages has been exponential for years. The packages can be used for almost any conceivable analytical problem, from X-ray image analysis to the classification of sound files. The packages are created by a worldwide community of high-ranking scientists and software developers. Large corporations also participate: IBM, Microsoft, Oracle, Google or Tibco are active promoters of the language. The power of this community is unparalleled by anything that a single company could create.

5 - The Toolset

Furthermore, the analyses can be flexibly embedded in existing processes through interfaces to high-level languages (Java, C, C++, etc.), business applications (SAP, Salesforce, etc.) and databases of all major providers.

6 - The Analysis Methods

The Big Method Case instead of “One-Algo-Fits-All”

6 - The Analysis Methods

The quote by Paul Watzlawick "If you only have a hammer as a tool, you see a nail in every problem" also applies to Data Science.

Even if the opposite is sometimes claimed: there is no single method that can solve every problem.

The choice of analysis method should always be based on the problem. It is therefore important to have a large method case available, from which the right tool can be taken.

6 - The Analysis Methods

Often even combinations of different methods, for example in the form of ensemble models or multi-level analyses, are more successful than individual methods.

6 - The Analysis Methods

The choice of methods should also take into account the target group of the analyses. For example, if sales planning is supported by Data Science, a minimum level of acceptance must be established within the department. This can best be achieved by a transparent and comprehensible approach, which is easier to achieve with some analysis methods than with others.

7 - The Industry Reference

Transfer instead of Island Thinking

7 - The Industry Reference

Data Science is always **embedded** in an **industry** and **company context**.

Nevertheless, the focus on the own industry should not be too strong. After all, another successful concept of Data Science is industry **transfer**. It is worth looking **beyond the horizon** - methods that were originally developed for other sectors can often be transferred to one's own use cases.

8 - The Expectations

Recognize Opportunities, but do not expect Miracles

8 - The Expectations

Data Science offers companies great opportunities to recognize connections and to forecast the future. At the same time it is important to recognize the limits of what is possible.

Data quality plays a role in this context, as does embedding.

8 - The Expectations

Many procedures are hypothesis-free, i.e. it is not necessary to formulate assumptions about relationships in advance.

This does not mean, however, that any data can be passed out-of-the-box to an algorithm, which then answers questions that have not yet been asked.

9 - The Course of the Project

Acceptance for a Journey of Discovery

9 - The Course of the Project

Data Science is like a journey of discovery. However, the treasures to be found are not necessarily where you expected them to be in advance.

9 - The Course of the Project

In order to realize the potential of Data Science, acceptance and openness towards a **highly agile** approach is therefore required on the part of the management or the leading specialist department.

If this is not a given from the outset, this aspect can also be best achieved through **transparency** and **continuous communication**.

10 - The Project Facility

Think Big - Start Small

10 - The Project Facility

Especially for companies taking their first steps in the field of Data Science, the project facility is elementary.

10 - The Project Facility

Data Science projects typically run along three levels of maturity:

1. Definition and evaluation of potential use cases.
2. Implementation of the most promising use cases as proof of concept.
3. Integration of successful proof-of-concept solutions into productive operation.

10 - The Project Facility

This multi-stage procedure is recommended in order to identify potentials and problems at an early stage.

The keyword is **Fail-Fast**.

10 - The Project Facility

With this strategy failures can be avoided and budgets and resources can be used effectively. Agile project management leads to faster successes and enables opportunities and risks to be weighed up on an evidence-based basis.

11 - The Departments

Take Along and Integrate

11 - The Departments

It is not unusual for departments to be sceptical about Data Science.

The content experts fear that Data Science, machine learning and artificial intelligence will cause them to lose tasks and esteem and therefore react cautiously or even negatively.

11 - The Departments

Dissolving or preventing this reluctance and establishing acceptance is a necessary condition for the success of Data Science projects.

Two aspects are central to this:

1. It must be made clear that the department will retain the leading role. It is about developing assistance systems that help to make better decisions. The development of autonomous algorithmic systems is neither realistic nor goal-oriented in most contexts.
2. The department should be motivated to support Data Science with expert competence in content. Both sides, Data Science and specialist departments, contribute to the success of a common goal.

12 Communication

Comprehensible Messages and Empathy instead of Statistical Gibberish

12 Communication

In addition to the technical task, the data scientists have the task of developing an understanding for the challenges and questions of the specialist departments and of explaining complex statistical procedures in a comprehensible way. It is important to avoid the use of technical statistical terms in relation to the specialist departments. It is better to present the procedure in a comprehensible way and thus demonstrate the benefits for all those involved.

12 Communication

Here the communication culture in the company plays a decisive role.

When employees work in an open, cooperative and team-oriented manner and transparency is rewarded, everyone involved ultimately benefits.

Summary

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